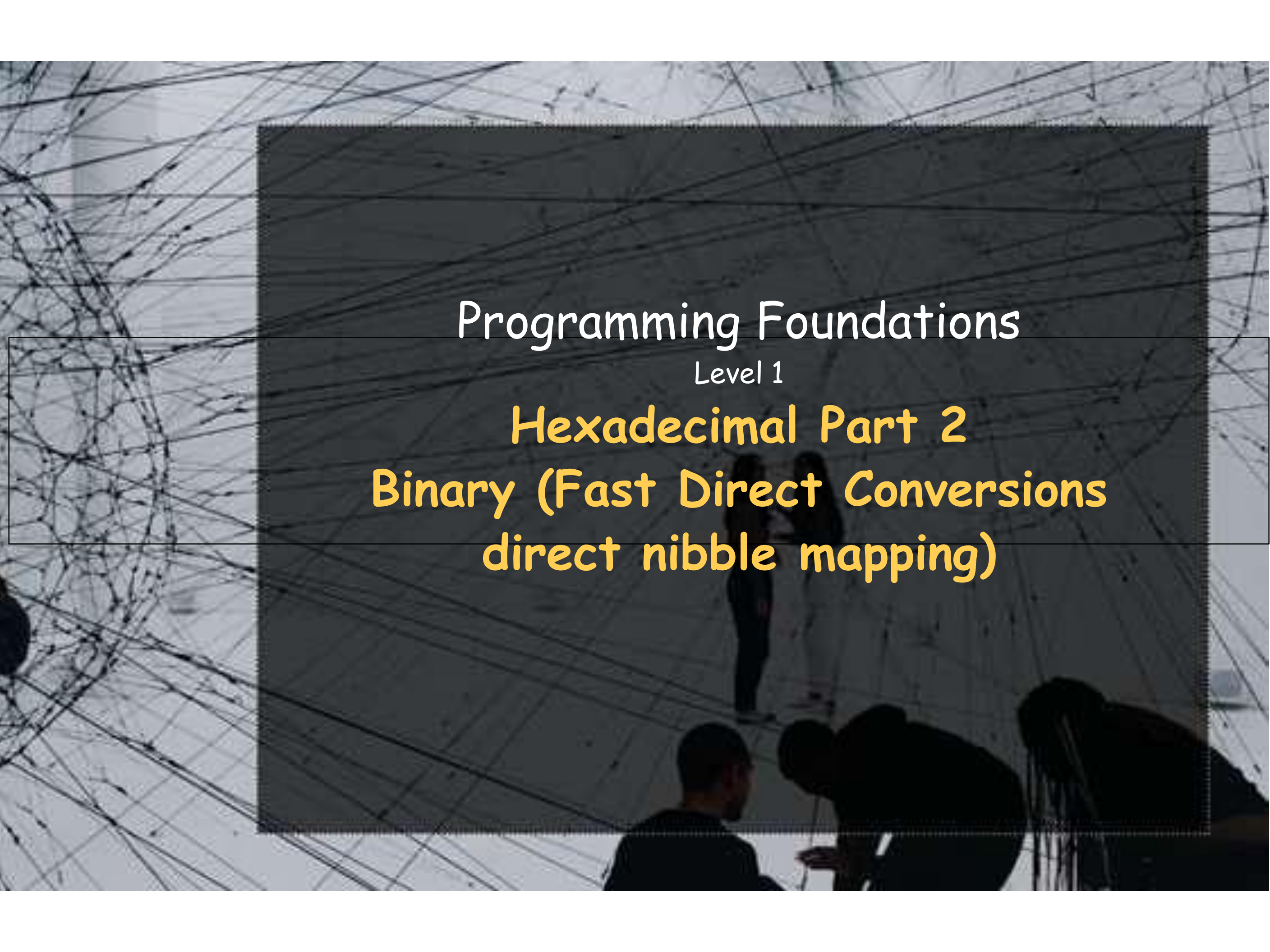




Programming Foundations Level 1



Programming Foundations

Level 1

Hexadecimal Part 2

Binary (Fast Direct Conversions direct nibble mapping)



Programming Foundations

Level 1

Convert Hexa to Binary?

Hexadecimal $\leftrightarrow$ Binary (Fast Direct Conversions - direct nibble mapping)

Hex is the most readable way to look at binary. In this lesson you'll learn **two ways** to convert between **hexadecimal** and **binary** — including the **direct nibble mapping** that turns every hex digit into **exactly 4 bits** (and back) with no arithmetic.

How to Convert Hexa to Binary?

Way 1 (Two-step):

Two steps:

1. Convert hexa to decimal.
2. Convert decimal to binary.

That's it.

How to Convert Hexa to Binary?

Way 2 (Direct):

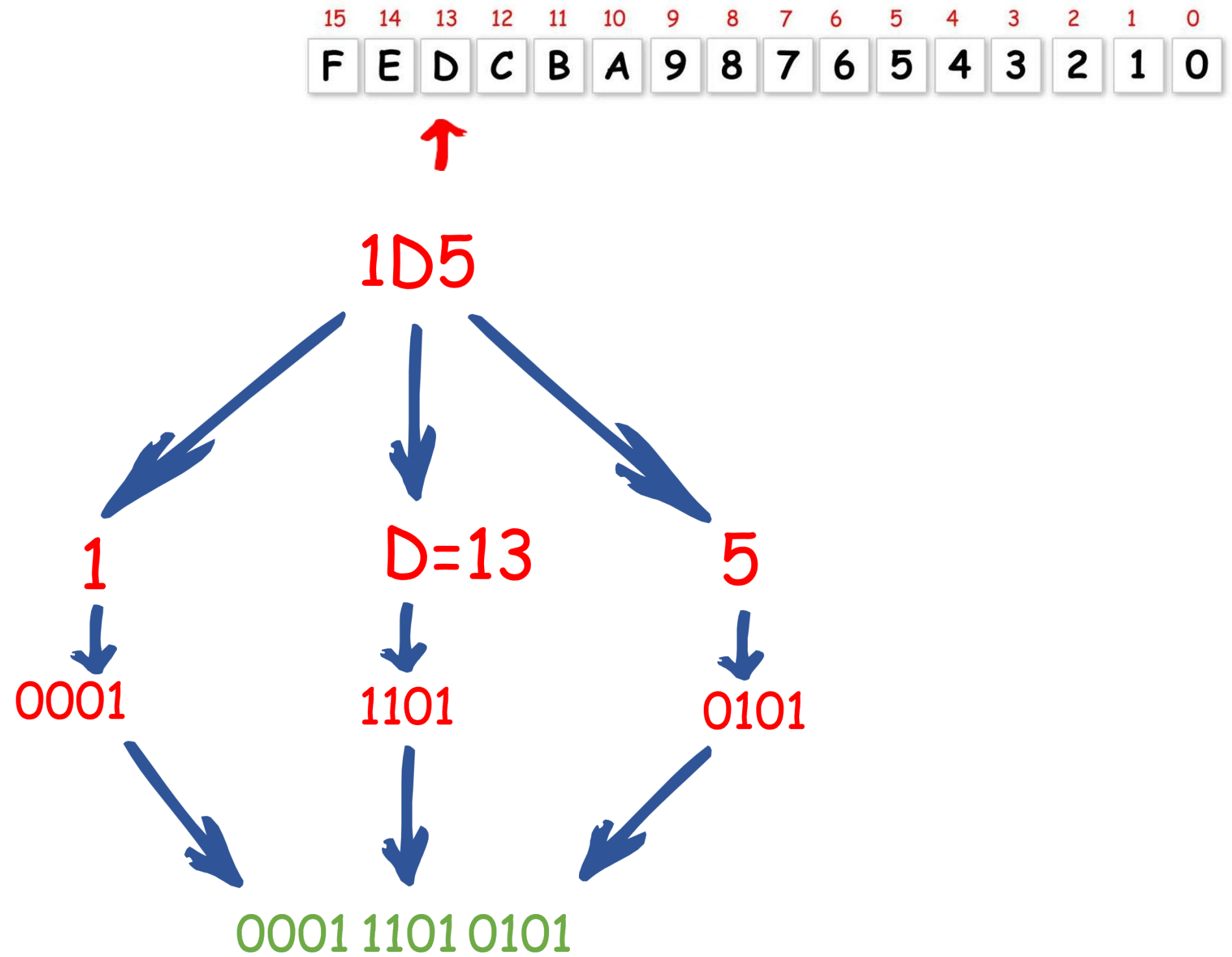
Map each hex digit $\rightarrow$ 4-bit nibble. (*Fastest; what pros use*).

Example mapping:

A $\rightarrow$ 1010, B $\rightarrow$ 1011, C $\rightarrow$ 1100, D $\rightarrow$ 1101, E $\rightarrow$ 1110, F $\rightarrow$ 1111.

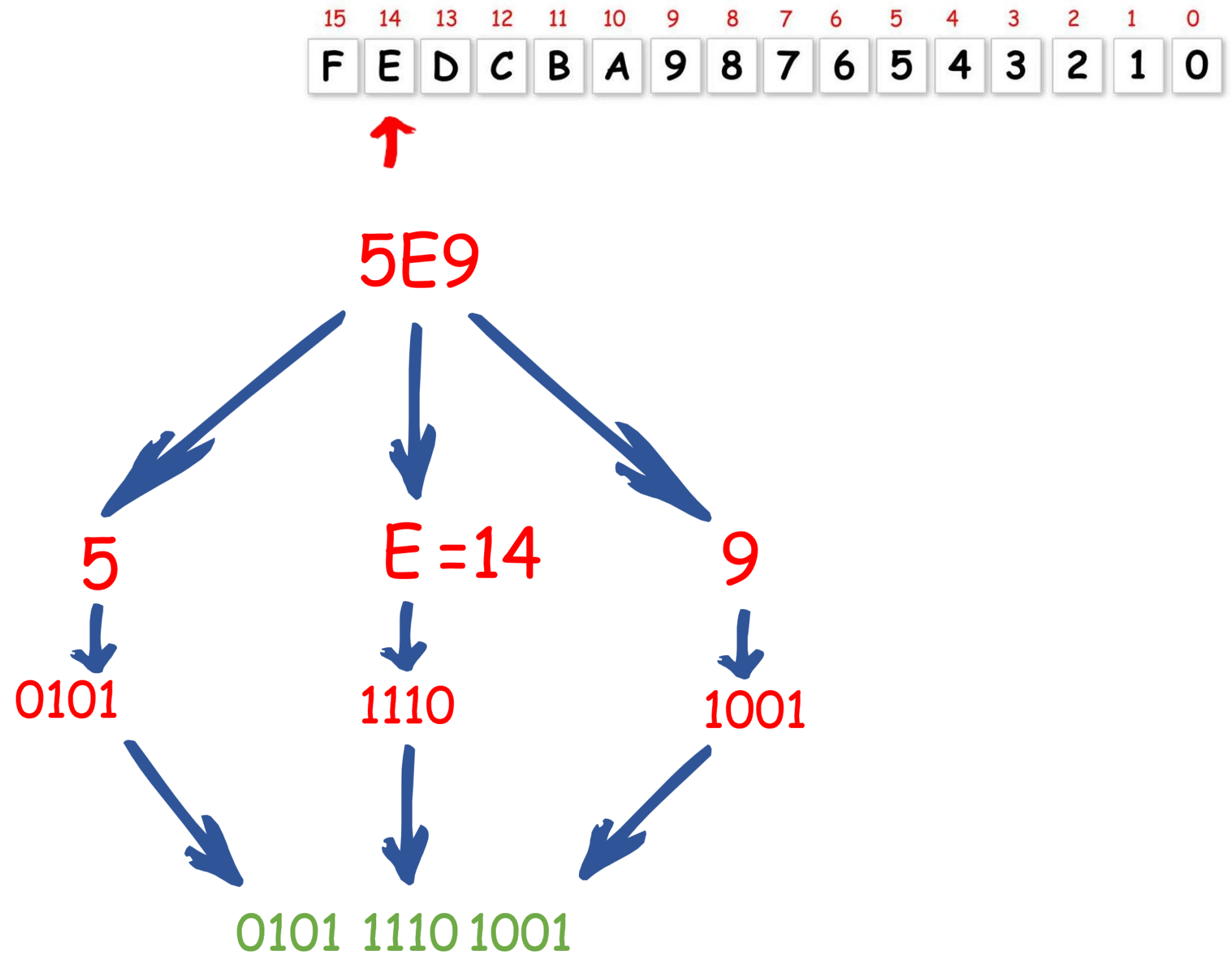
Direct Conversion:

Convert **1D5** to Binary



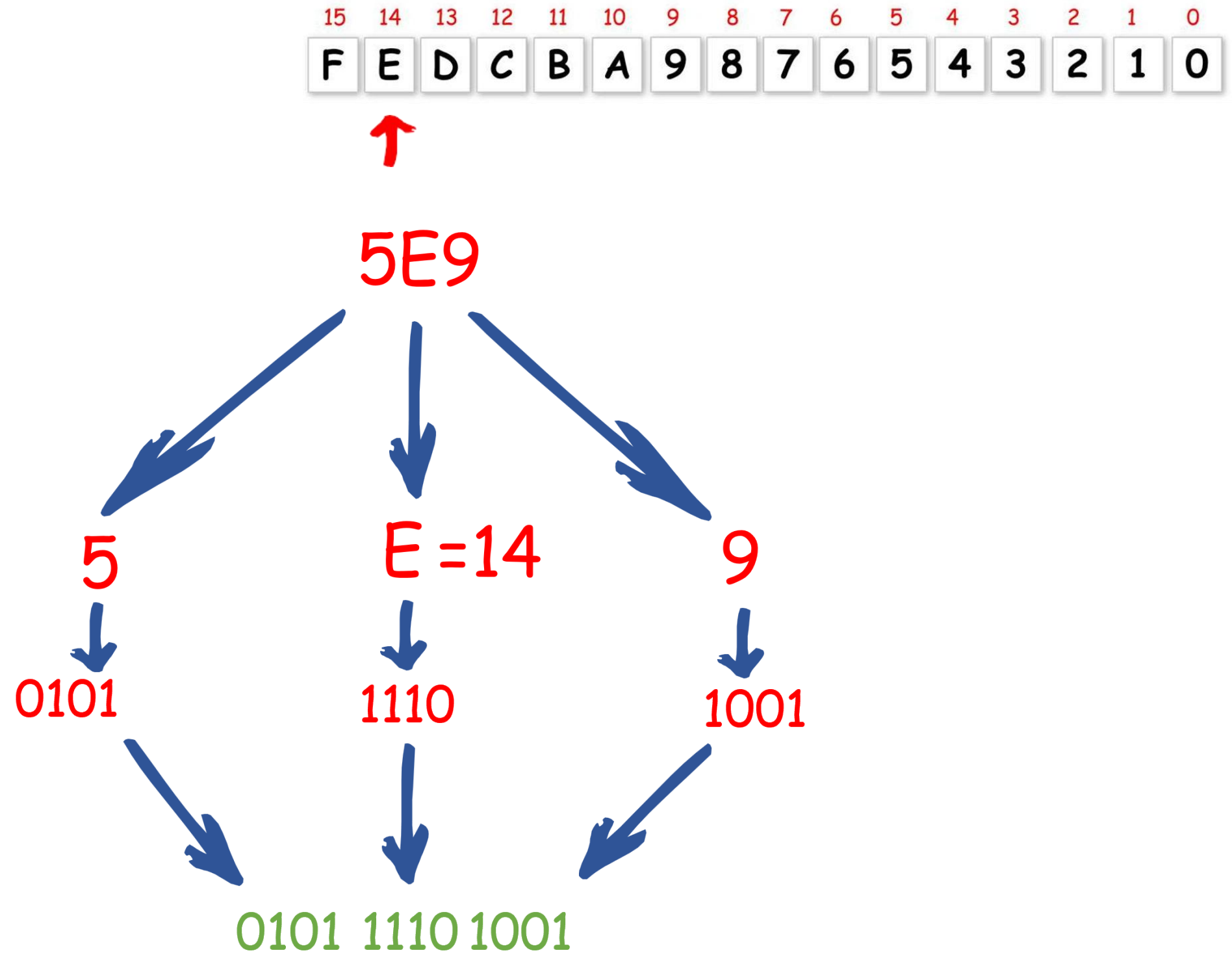
Direct Conversion:

Convert **5E9** to Binary



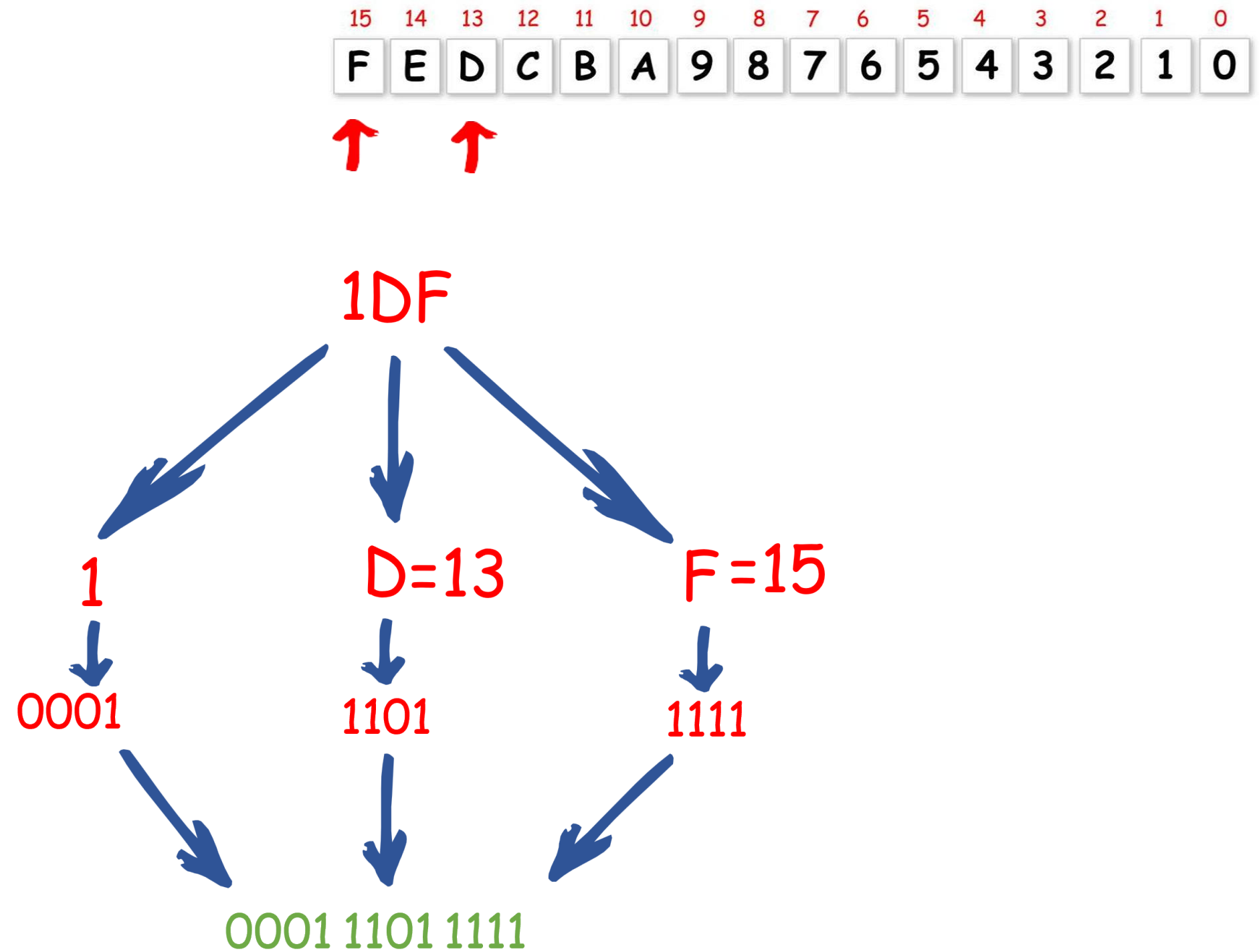
Direct Conversion:

Convert **5E9** to Binary



Direct Conversion:

Convert **1DF** to Binary



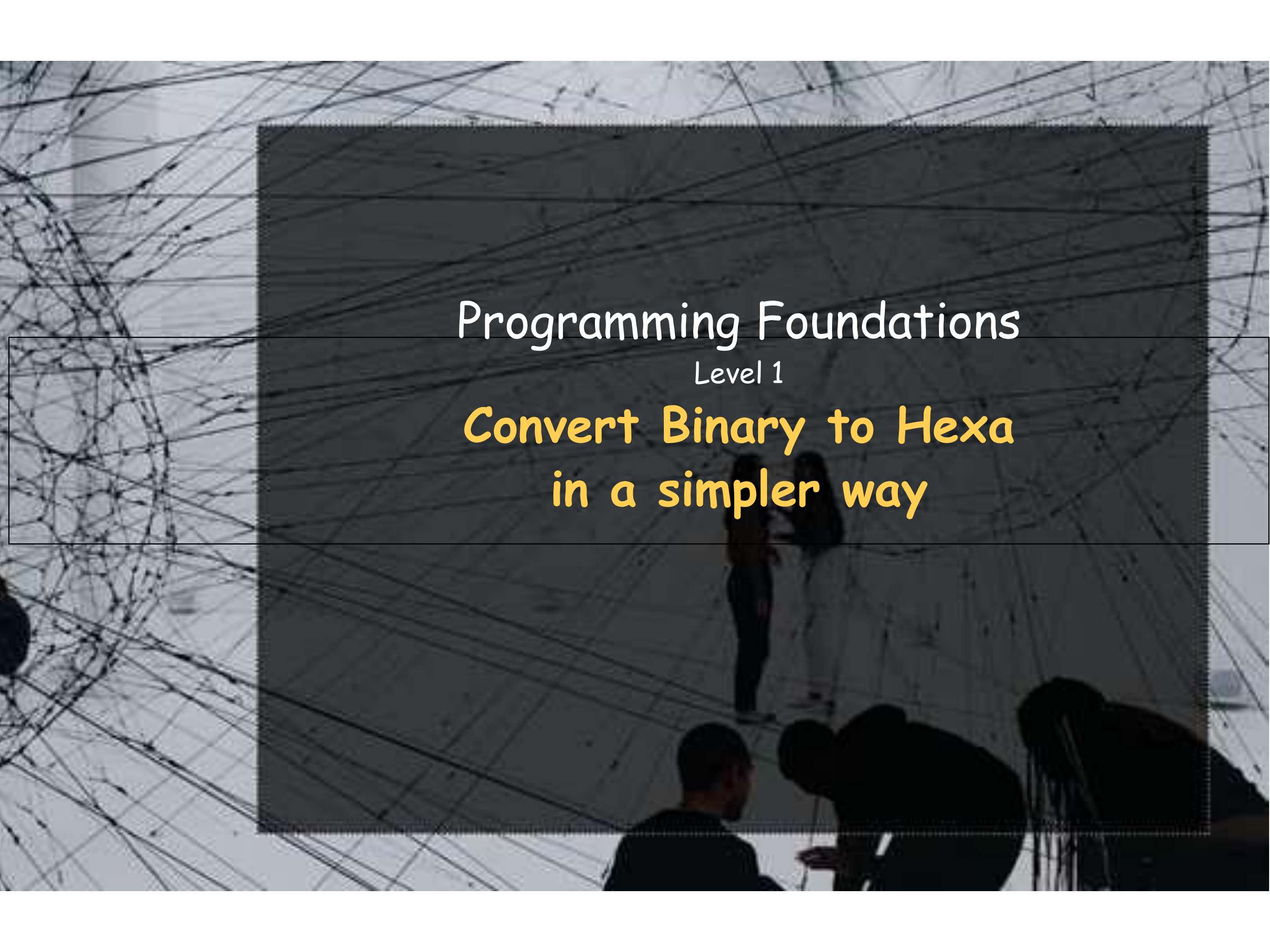
Worked examples:

■ $1D5_{16} \rightarrow 1 \mid D \mid 5 \rightarrow 0001\ 1101\ 0101 \rightarrow \mathbf{0001\ 1101\ 0101}.$

■ $5E9_{16} \rightarrow 5 \mid E \mid 9 \rightarrow 0101\ 1110\ 1001 \rightarrow \mathbf{0101\ 1110\ 1001}.$

■ $1DF_{16} \rightarrow 1 \mid D \mid F \rightarrow 0001\ 1101\ 1111 \rightarrow \mathbf{0001\ 1101\ 1111}.$

Tip: Always group the final binary as **nibbles (4 bits)** for readability.



Programming Foundations

Level 1

**Convert Binary to Hexa
in a simpler way**

How to Convert Binary to Hexa?

Way 1 (Two-step):

Two steps:

1. Convert binary to decimal.
2. Convert decimal to hexa.

That's it.

How to Convert Binary to Hexa?

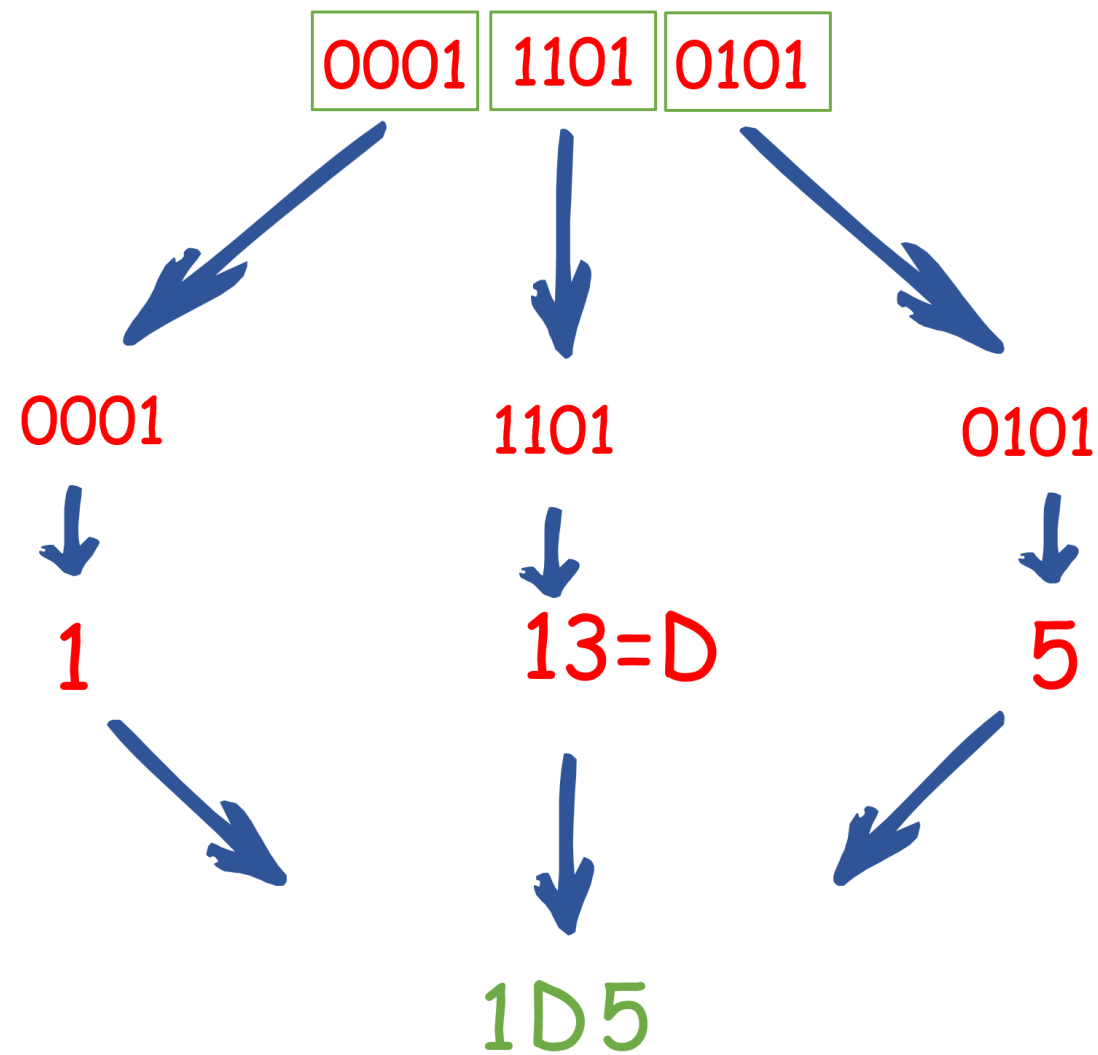
Way 2 (Direct):

Group bits in fours from the right, then map each 4-bit group to one hex digit.

Direct Conversion:

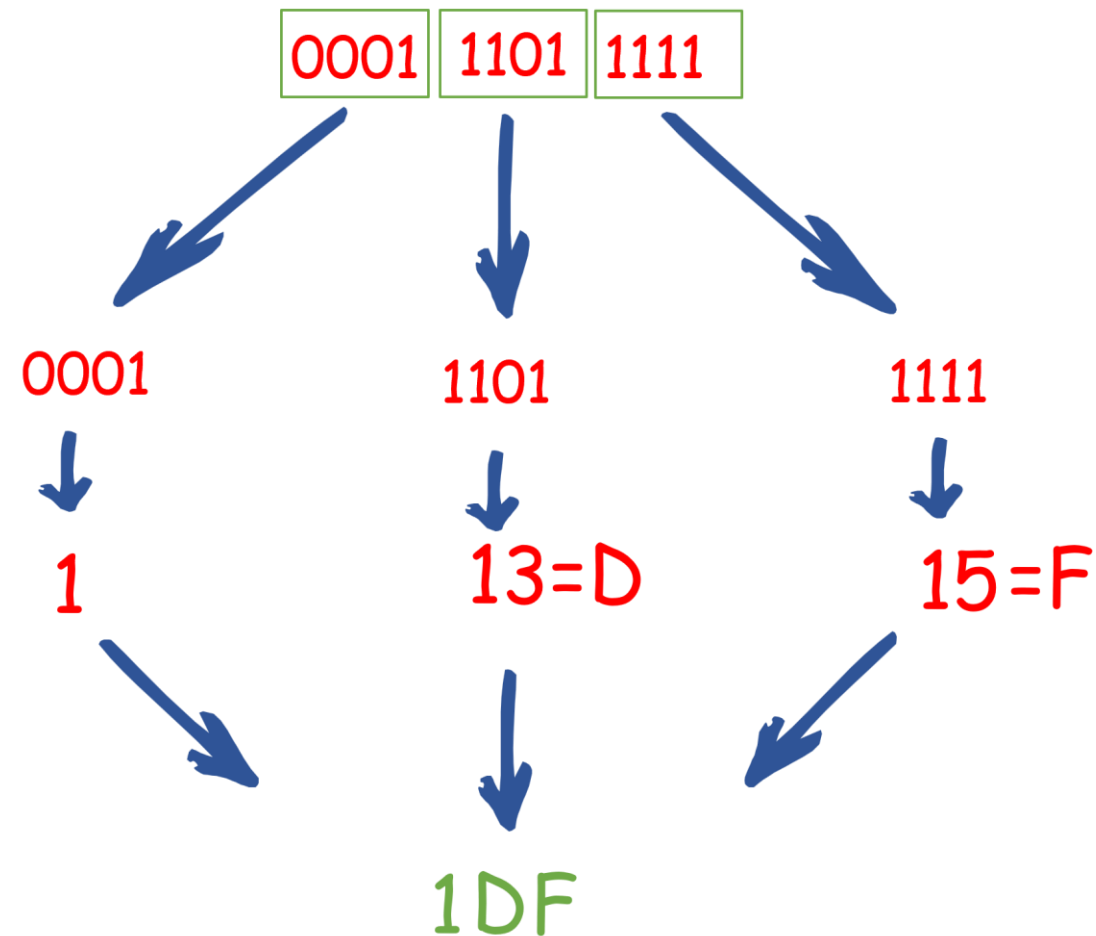
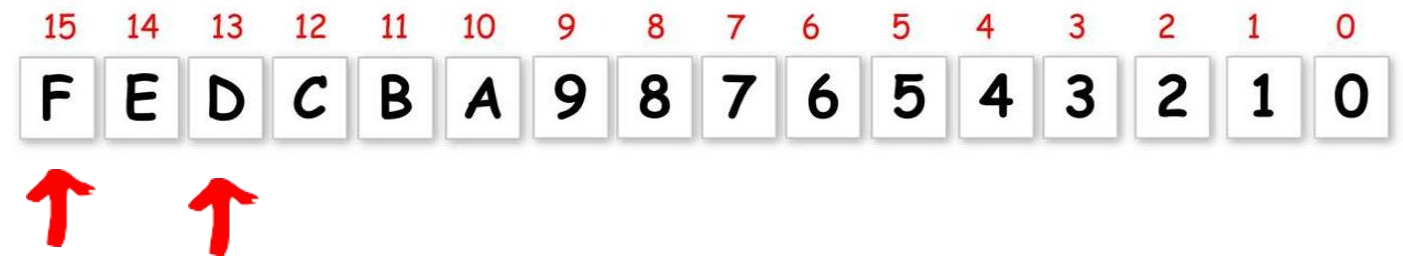
Convert **000111010101** to Binary

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0



Direct Conversion:

Convert **000111011111** to Binary



Worked examples:

0001 1101 0101₂ → group 0001 | 1101 | 0101 → **1 D 5** → **1D5₁₆**.

0001 1101 1111₂ → group 0001 | 1101 | 1111 → **1 D F** → **1DF₁₆**.

Why Direct Conversion Wins

- Zero arithmetic once you know the nibble table.
- One byte = two hex digits → memory dumps, instruction bytes, and color codes are clean and short.

Interconnection

- ◆ **Hex ↔ Binary** maps 1 digit ↔ 4 bits (nibble), so 1 byte ↔ 2 hex digits.
- ◆ Use **Way 2 (direct)** for speed; use **Way 1** to reinforce base-conversion theory.
- ◆ Grouping in **nibbles** keeps numbers readable and reduces errors when reading bytes, addresses, and encodings.

Activity

Instructions:

Convert **directly** (use nibble mapping / 4-bit grouping only).

1 Hex → Binary

- a) 12A4
- b) 1C35
- c) 100
- d) 115C

2 Binary → Hex

- a) 0001 0010 1010 0100
- b) 0001 1100 0011 0101
- c) 0001 0000 0000
- d) 0001 0001 0101 1100



Activity Solution



Solution:

1 Hex → Binary

- a) **12A4** → 0001 0010 1010 0100
- b) **1C35** → 0001 1100 0011 0101
- c) **100** → 0001 0000 0000
- d) **115C** → 0001 0001 0101 1100

2 Binary → Hex

- a) 0001 0010 1010 0100 → **12A4**
- b) 0001 1100 0011 0101 → **1C35**
- c) 0001 0000 0000 → **100**
- d) 0001 0001 0101 1100 → **115C**

Conclusion

- ✓ **Direct nibble mapping** is the professional, fast path for **Hex** ↔ **Binary**.
- ✓ Keep everything **grouped in 4-bit nibbles**, pad with zeros on the left when needed.
- ✓ Use **two-step** (via decimal) to validate learning or when teaching base arithmetic foundations.

With these skills, your students can confidently read bytes, registers, addresses, and encodings in real-world tools and labs. 🚀

Thank you 😊